

CS 577 - Introduction to Algorithms

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Fall 2026



CS 577 - INTRODUCTION TO ALGORITHMS: FALL 2026

ANALYSIS OF ALGORITHMS

Problem

- Mathematical model of the problem area.
- Rules of the game.

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- Ex: I have kitchen with a stocked pantry and I want a cookie.

Algorithm

- Step-by-step procedure for solving an *instance* of a given problem.
-
- Ex: Given a kitchen with a stove, etc... and a pantry with chocolate chips, etc...

Chocolate Chip Cookies

Ingredients:

- 227g (1 cup) butter, softened
- 200g (1 cup) sugar
- 105g (½ cup) brown sugar
- 2 eggs
- 2 tsp vanilla
- 250g (2 cups) all-purpose flour
- 1 tsp soda
- 1 pinch salt
- 1 ½ cups of chocolate chips

Instructions:

1. Beat butter, sugars, eggs and vanilla until light and fluffy.
2. Add flour, soda, and salt; blend well.
3. Add chips.
4. Drop from a teaspoon 2 inches apart.
5. Bake 190°C for 9 min.

STABLE MARRIAGE PROBLEM (SMP) (1962)¹²³

Problem Definition

Given a set of n men, M , and an opposite set of n women, W . Each person has a preference ranking of the opposite set. Compute a stable matching between M and W . A matching is stable if it is (i) perfect, and (ii) there are no pairs (m, w) and (m', w') in the matching where m prefers w' and w' prefers m .

¹Algorithm Design, Ch 1.

²Algorithms, Ch 4.5

³<http://mathsite.math.berkeley.edu/smp/smp.html> (Uses Flash)

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- A.k.a Stable Matching Problem.
- There are more complicated variations of the model.
- Used in the real world (e.g. matching doctors to hospitals).
- Nobel Prize in Economics in 2012 (Shapley and Roth).

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GALE-SHAPELY ALGORITHM⁴ FOR SMP (1962)

INITIALLY ALL $m \in M$ AND $w \in W$ ARE FREE

while *there is a man m who is free and hasn't proposed to every woman* **do**

 CHOOSE SUCH A MAN m

 LET w BE THE HIGHEST-RANKED WOMAN IN m 'S PREFERENCE LIST TO WHOM m HAS NOT YET PROPOSED

if w *is free* **then**

(m, w) BECOME ENGAGED

else w *IS CURRENTLY ENGAGED TO m'*

if w *prefers m' to m* **then**

m REMAINS FREE

else w *PREFERS m TO m'*

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end

end

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return *the set S of engaged pairs*

⁴Algorithm Design, p.6

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- Complete?

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- Correct?

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return *the set S of engaged pairs*

Is it good?

- Complete?
- Correct?
- Efficient? With respect to what (time, space, ...)?

⁴Algorithm Design, p.6

ABOUT YOU

My current year in school is:

- a. Freshman
- b. Sophomore
- c. Junior
- d. Senior
- e. Graduate Student
- f. Other

ABOUT YOU

My primary reason for taking CS 577:

- a. I am very interested in the subject.
- b. I am curious to learn more about the subject.
- c. It fulfils a requirement for my program, major or certificate.
- d. It fits my schedule.
- e. I've heard good things about the course.

ABOUT YOU

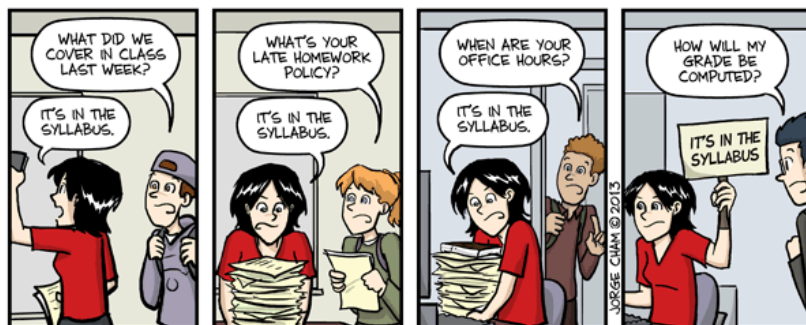
My favourite Star Wars movie (from the trilogies) is:

- a. I - The Phantom Menace
- b. II - Attack of the Clones
- c. III - Revenge of the Sith
- d. IV - A New Hope
- e. V - The Empire Strikes Back
- f. VI - Return of the Jedi
- g. VII - The Force Awakens
- h. VIII - The Last Jedi
- i. IX - The Rise of Skywalker
- j. Never seen them

COURSE OVERVIEW

SYLLABUS (COURSE LOGISTICS)

[HTTPS://CANVAS.WISC.EDU/COURSES/505881](https://canvas.wisc.edu/courses/505881)

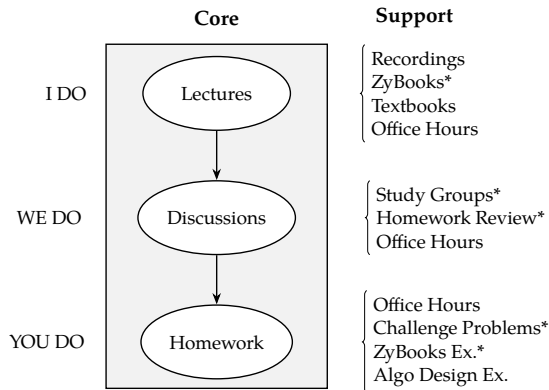


IT'S IN THE SYLLABUS

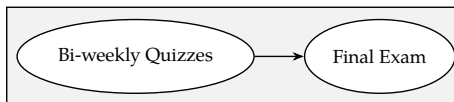
This message brought to you by every instructor that ever lived.

WWW.PHDCOMICS.COM

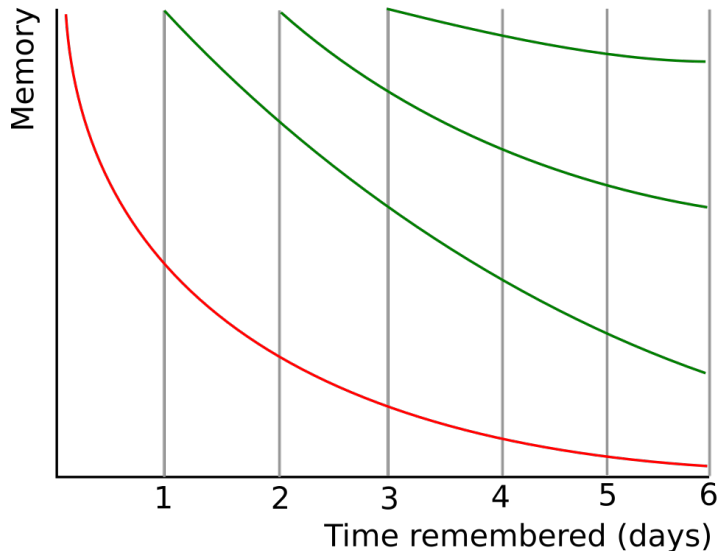
CS577 Intro to Algorithms Overview



Assessments



The Forgetting Curve



REVIEW THE SYLLABUS

Grading

- Participation (25%)
 - TopHat Questions (5%)

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 - TopHat Questions (5%)
 - Discussion Participation (5%)

REVIEW THE SYLLABUS

Grading

- Participation (25%)
 - TopHat Questions (5%)
 - Discussion Participation (5%)
 - Assignments (15%)
 - Graded on participation not correctness!
 - Participation credit requires a reasonable attempt to answer a question.
 - Copying answers, or using an AI is not valid for participation marks and will result in 0 for entire assignment. This is considered plagiarism which can result in disciplinary action.

REVIEW THE SYLLABUS

Grading

- Participation (25%)
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 - Discussion Participation (5%)
 - Assignments (15%)
 - All have the 80% rule

REVIEW THE SYLLABUS

Grading

- Participation (25%)
 - TopHat Questions (5%)
 - Discussion Participation (5%)
 - Assignments (15%)
 - All have the 80% rule
- Quizzes (30%) [See syllabus for dates.]
 - 30 minute Canvas quiz done in the new testing facility.
 - Let a be an array (0 based indexing) containing your 6 quiz scores, in order, from highest to lowest.

$$\text{Quizzes score} = a[0] \cdot 15 + a[1] \cdot 8 + a[2] \cdot 4 + a[3] \cdot 2 + a[4] \cdot 1 + a[5] \cdot 0$$

REVIEW THE SYLLABUS

Grading

- Participation (25%)
 - TopHat Questions (5%)
 - Discussion Participation (5%)
 - Assignments (15%)
 - All have the 80% rule
- Quizzes (30%) [See syllabus for dates.]
 - 30 minute Canvas quiz done in the new testing facility.
- Exam (45%)
 - Monday, Dec 14, 2026 @ 17:05 to 19:05 (45%)

REVIEW THE SYLLABUS

Flexibility Built-in for Everyone

- 80% rule for Participation.
- Generous quiz aggregation.

Course Expectations

Doing less than 80% of the assigned discussions, classes, and assignments risks altering the knowledge and skills of the course, lowering the academic standards, and fundamental altering the nature of the course.

- We expect every student to attend lectures, discussions, submit all homework, and do all the quizzes.
- The flexibility is provide because life happens NOT because we expect students to only do 80% of the work or skip quizzes.

REVIEW THE SYLLABUS

Bonus Points (Capped at 10%)

- Bonus calculations are as is: No extensions, flexibility, or exceptions.
- Syllabus Canvas Quiz (2%)
- PreReq Review Assignment (2%)
- In-Person Lecture Attendance (5%):
 - Determined by TopHat attendance function; 80% rule applied.
 - For credit, you need to be in class and have your tech working.
- Challenge Problems (5%):
 - 6 exam like questions release throughout the semester.
 - 1 hour at drop-in offices hours.
 - 50% from “live solving”, and 50% from submitted solution.

REVIEW THE SYLLABUS

Bonus Points (Capped at 10%)

- Attending one study group per week (5%)
- Homework Review (5%):
 - Once the solutions are released, review your solutions and discuss with instructor, TA, or PM in office hours.
 - Come prepared!
 - If office hours are busy, you may be asked to come back at a later time.
- zyBook (5%):
 - 2% for participation activities.
 - 3% for designated exercises (see assignments on Canvas).

REVIEW THE SYLLABUS

Academic Integrity

- Academic dishonesty or misconduct is taken very seriously by the university (see UW–Madison Academic Integrity policy).
- It is academic misconduct to submit someone else's work as your own.
- It is academic misconduct to help another student commit academic misconduct.

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- It is academic misconduct to help another student commit academic misconduct.
- Submissions that copy solutions from peers, solutions found online, or chatbots/LLMs will be given a 0.

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Peer Help on Assignments

- You may not email, post on Piazza, or otherwise make solutions (or part of) available for others.
- Process:
 - If you receive or give help on an assignment, be sure to cite them.

DON'T USE CHATBOTS FOR THIS COURSE

MIT study: Using ChatCPT prevents learning and hurts your cognitive capacity (makes you dumber)

<https://arxiv.org/pdf/2506.08872>

- 83.3% of ChatGPT users couldn't quote from essays they wrote minutes earlier vs about 15% who did not use ChatGPT.
- Brain scans revealed the damage: neural connections collapsed from 79 to just 42.
- When researchers forced ChatGPT users to write without AI, they performed worse than people who never used AI at all.
- More highlights: <https://x.com/itsalexvacca/status/1935343874421178762>

DON'T USE CHATBOTS FOR THIS COURSE

MIT report: The GenAI Divide: State of AI in Business 2025

https://mlq.ai/media/quarterly_decks/v0.1_State_of_AI_in_Business_2025_Report.pdf

- “Despite [large] investments in GenAI, ... 95% of organizations are getting zero return.”
- “[GenAI] primarily enhance individual productivity, not P&L performance.”

GETTING HELP

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[HTTPS://CANVAS.WISC.EDU/COURSES/505881](https://canvas.wisc.edu/courses/505881)

Help!

- Piazza Online Discussion
- Weekly Discussions
- Weekly Study Groups on Specific Topics
(Watch Piazza for sign-ups)
- TA Office Hours
- Instructor Office Hours



APPENDIX

REFERENCES

IMAGE SOURCES I

TOP HAT

<https://tophat.com/>

piazza

<https://piazza.com/>



WISCONSIN
UNIVERSITY OF WISCONSIN-MADISON

<https://brand.wisc.edu/web/logos/>



[http://bigpicture.typepad.com/comments/
images/2008/07/14/dont_panic.png](http://bigpicture.typepad.com/comments/images/2008/07/14/dont_panic.png)



IT'S IN THE SYLLABUS
No technology brought to you by the author. Read and learn.

<http://phdcomics.com/comics.php?f=1583>

IMAGE SOURCES II



[https:](https://www.linkedin.com/company/gradescope/)

[//www.linkedin.com/company/gradescope/](https://www.linkedin.com/company/gradescope/)



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